

INFORMATION STORAGE AND RETRIEVAL (Professional Elective - III)

Course	B.Tech.-VII-Sem.	L	T	P	C
Subject Code	22CSPE72	3	-	-	3

Course Outcomes (COs) & CO-PO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	Upon completion of course the students will be able to	PO2	PO3	PO5	PO6	PO12	PSO1
CO1	outline the importance of Information storage and Retrieval	3	3	3	3	3	3
CO2	illustrate cataloging and indexing in information storage	3	2	3	3	3	3
CO3	adapt automatic indexing and clustering in information storage	3	3	3	3	3	3
CO4	implement user search techniques	3	3	3	3	3	3
CO5	apply text search algorithm in information retrieval	3	2	2	3	3	3

Syllabus

Unit	Title/Topics	Hours
I	Introduction	8
Introduction: Definition, Objectives, Functional Overview, Relationship to DBMS, Digital libraries and Data Warehouses. Information Retrieval System Capabilities: Search, Browse, Miscellaneous. Task: Perform a case study on Information Retrieval System Capabilities.		
II	Cataloging and Indexing	7
Objectives, Indexing Process, Automatic Indexing, Information Extraction. Data Structures: Introduction, Stemming Algorithms, Inverted file structures, N-gram data structure, PAT data structure, Signature file structure, Hypertext data structure. Task: Perform a case study on Hypertext data structure.		
III	Automatic Indexing and Clustering	9+8=17
Part-A: Automatic Indexing: Classes of automatic indexing, Statistical indexing, Natural language, Concept indexing, Hypertext linkages. Task: Perform a case study on Statistical indexing		
Part-B: Document and Term Clustering: Introduction, Thesaurus generation, Item clustering, Hierarchy of clusters. Task: Perform a case study on Hierarchy of clusters.		
IV	Search Techniques and Visualization	7
User Search Techniques: Search statements and binding, Similarity measures and ranking, Relevance feedback, Selective dissemination of information search, weighted searches of Boolean systems, Searching the Internet and hypertext. Information Visualization: Introduction, Cognition and perception, Information visualization technologies. Task: Perform a case study on Cognition and perception of human being.		
V		9
Text Search Algorithms: Introduction, Software text search algorithms, Hardware text search systems. Information System Evaluation: Introduction, Measures used in system evaluation, Measurement example - TREC results. Task: Perform a case study on Software text search algorithms.		
Textbooks		
1. Information Storage and Retrieval Systems: Theory and Implementation by Gerald J. Kowalski, Mark T. Maybury, Second Edition, Kluwer Academic Publishers. 2000		
References		
1. Frakes, W.B., Ricardo Baeza-Yates: Information Retrieval Data Structures and Algorithms, Prentice Hall, 1992.		
2. Modern Information Retrieval By Yates Pearson Education.		